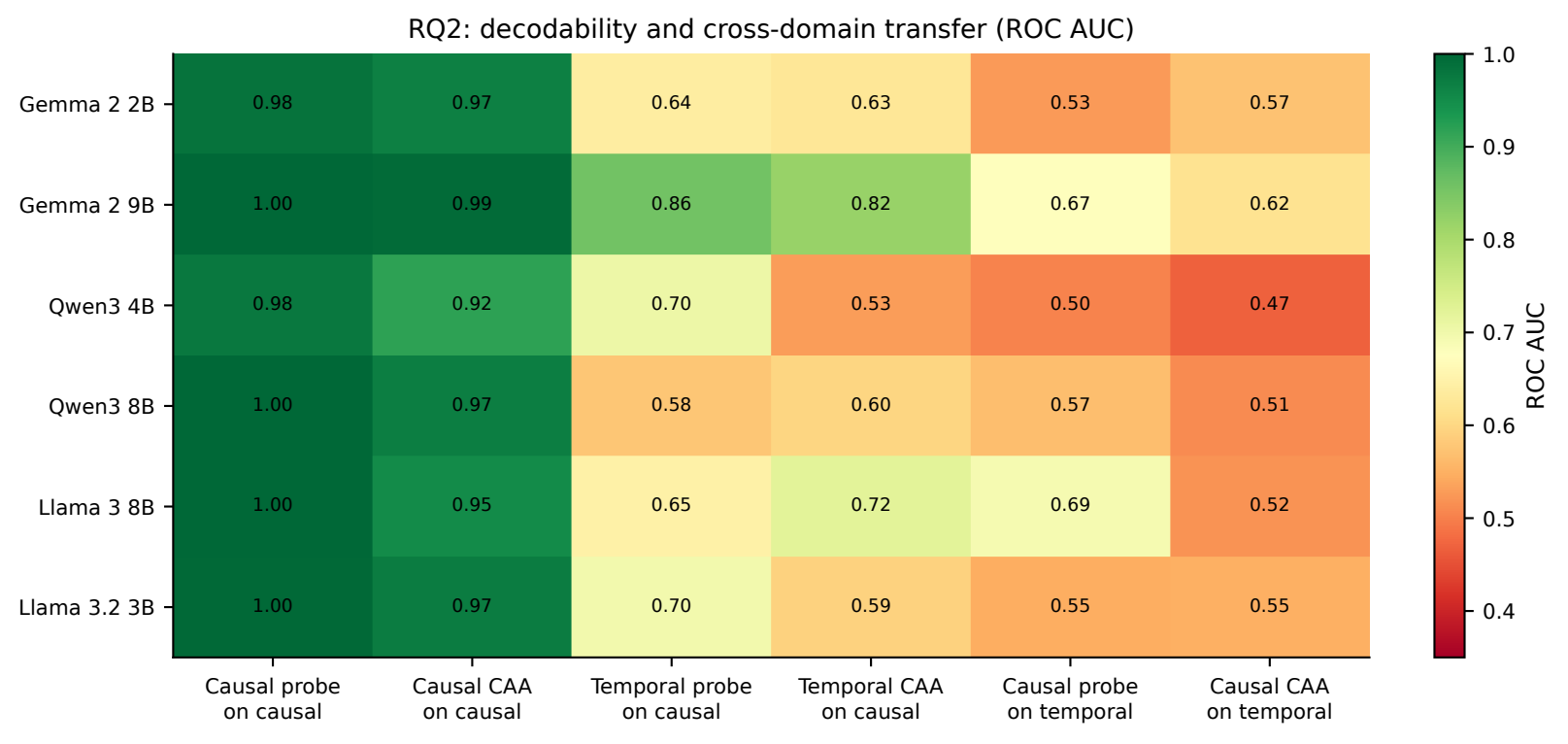
RQ2: decodability and cross-domain transfer (ROC AUC)
Gemma 2 2B
Gemma 2 9B
Qwen3 4B
Qwen3 8B
Llama 3 8B
Llama 3.2 3B
0.98
0.97
0.64
0.63
0.53
0.57
1.00
0.99
0.86
0.82
0.67
0.62
0.98
0.92
0.70
0.53
0.50
0.47
1.00
0.97
0.58
0.60
0.57
0.51
1.00
0.95
0.65
0.72
0.69
0.52
1.00
0.97
0.70
0.59
0.55
0.55
Causal probe on causal
Causal CAA on causal
Temporal probe on causal
Temporal CAA on causal
Causal probe on temporal
Causal CAA on temporal
ROC AUC
1.0
0.9
0.8
0.7
0.6
0.5
0.4